

OMMEY HABIBA SOWME

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SUMMARY

Accomplished Computer Science student with a 3.91 CGPA and published research experience in Machine Learning. Skilled in developing high-accuracy classification models, large-scale datasets, and secure cryptographic systems for web and mobile platforms.

EDUCATION

B.Sc. in Computer Science and Engineering: CGPA 3.91/4.00 [2022-2026]
R.P. Shaha University

HSC (Science): GPA 5.00/5.00 [2020]
Narayanganj Government Mohila College

SSC (Science): GPA 5.00/5.00 [2018]
Morgan Girls School and College, Narayanganj

PUBLICATIONS (SCOPUS INDEXED)

Ommey Habiba Sowme et al. "Visual Classification of Ten Region-Specific Flower Species in Bangladesh Using Hybrid CNN and Vision Transformers" on *IEEE 28th International Conference on Computer & Information Technology (ICCIT 2025)*

Status: Presented (Dec 2025) & Published (May) | DOI: [10.1109/ICCIT68739.2025.11490276](https://doi.org/10.1109/ICCIT68739.2025.11490276)

- Led the end-to-end development of a 23,000-image dataset for 10 native flower species, including manual field collection, labeling, and data augmentation to ensure model robustness.
- Developed and optimized various CNN and Vision Transformer architectures (ResNet, GoogLeNet, Swin, ViT), achieving up to 99% accuracy and engineering a hybrid ensemble model for balanced predictions.

THESIS & CORE PROJECTS

Proprietary Modification of Diffie-Hellman Public Parameters | Group Thesis Project

- Developed a revised DH methodology concealing generator (g) and prime (p) parameters to mitigate MitM and precomputation attacks.
- Implemented a 16-round Feistel Network cipher with dynamic timestamps in key scheduling to ensure session-level ciphertext variability.
- Designed a Beta (B) Wrapper using XOR Shift for secure, non-raw credential distribution over public Internet channels.

Cat Sound Emotion Classifier (96% Accuracy)

- Developed a 2D CNN architecture to classify 10 distinct feline vocalizations (Happy, Pain, Hunting, etc)
- Achieved 96% test accuracy through optimized feature extraction of Mel-Spectrograms.
- Deployed a functional mobile & web application for real-time audio inference.

RESEARCH INTERESTS

Computer Vision • Deep Learning • Vision Transformers (ViT) • Cryptography & Network Security

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL

Frameworks: PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, PyCryptodome, librosa

Tools: Git, LaTeX, VS Code, Google Colab

Other Skills: Research writing, Data preprocessing, Model evaluation, Cryptography (Basic Concepts & Implementation)

HONORS & REFERENCES

Academic Excellence Award: For top-tier CGPA throughout 8 semesters.